LECTURE 4

SUMMARY

MODULE 1 –LECTURE 4

UNDERSTANDING YOUR BODY'S ABILITY TO LOSE FAT & GAIN MUSCLE

HOW TO ACTUALLY DO A BODY ASSESSMENT (ACCURATE ESTIMATION METHODS)

Body assessment is the foundation of any transformation journey.
If you cannot accurately estimate:

- your body fat
- your maintenance calories
- your BMR
- your daily fluctuations

...then your transformation planning (bulking/cutting) will be incorrect.

This lecture explains exactly how to assess your body, calculate averages, and find true maintenance.

HOW DO WE ESTIMATE BODY FAT % AND ASSESS THE BODY?

If someone does not know how to estimate body fat or assess progress, here are the valid methods:

1) Professional Guidance

If a coach has coached thousands of clients, they develop an eye for estimating:

body fat %

body shape

fat distribution

water retention

muscle fullness

A trained eye can estimate fairly accurately by looking at progress photos.

2) DEXA Scan

Gold standard for body fat %, muscle mass, bone density.
Highly accurate but expensive.

3) Body Composition Analysis (BIA Machines)

Portable, accessible, but less accurate.
Useful for month-to-month comparison, not for exact numbers.

4) Skinfold Calipers

Very accurate when used by a skilled professional.
Measures subcutaneous fat from key sites.

5) Navy Method

Uses simple circumference measurements and online calculators.
Useful when no equipment is available.

UNDERSTANDING THE SCALE (DAILY WEIGHT FLUCTUATION)

A scale does NOT show:

- fat gain/loss
- muscle gain/loss
- water retention vs inflammation

It only shows total body weight, which changes daily based on:

- sodium intake
- carbohydrate intake (glycogen + water)
- hydration
- sleep quality
- stress levels
- digestion / bloating
- menstrual cycle
- intensity of previous day's training

That's why daily readings are useless alone— we only track weekly averages.

HOW TO CALCULATE WEEKLY AVERAGES

You track daily:

previous day's calories

next morning weight

cardio

workout intensity (1–10)

sleep quality (1–10)

sodium

digestion

Then you find:

Week 1 average weight
Week 2 average weight

Example:

Week 1 avg = 86.8 kg

Week 2 avg = 87.4 kg

87.4 – 86.8 = +0.6 kg

→ weight gain → you are in surplus
→ your calorie estimate is wrong → maintenance lower than you thought

Same logic works for weight loss.

HOW TO CALCULATE ACTIVE DEFICIT OR SURPLUS

If you lost weight:

Weight lost × 7700 = total deficit (calories)
7700 kcal = approx. 1 kg fat

For example:

You lost 0.5 kg over 14 days:

(0.5 × 7700) / 14 = 275 calories/day deficit

If you were eating 2000 cals per day:

Maintenance = 2000 + 275 = 2275 calories

This is your exact maintenance, not any formula.

Knowing maintenance calories is the most important step.

UNDERSTANDING HOMEOSTASIS

Your body ALWAYS tries to maintain balance.
The point where you neither gain nor lose weight is:

Homeostatic Point = true maintenance

When your body is stable, digestion normal, hunger normal, weight steady—that is your real maintenance.

You must be aware of your habits daily to keep the body stable.

HOW TO CALCULATE BMR (BASAL METABOLIC RATE)

Common quick estimation method:

Bodyweight (kg) × 22 = BMR
But this is still an estimate.

More accurate method using MET (Metabolic Equivalent)

BMR formula:
3.5 × LBM(kg) ÷ 200 × 60 × 24

(Your note had a small formatting error; this is the corrected logic.)

MET values:

Sleeping = 0.8–1.0 MET

Exercise = 3.5–10 MET depending on intensity

Lean Body Mass Calculation

LBM = Bodyweight – Fat mass

Example:
Bodyweight = 87 kg
Body fat = 15%
Fat mass = 87 × 0.15 = 13.05 kg
LBM = 87 – 13.05 = ~74 kg

LBM is used because fat does not burn many calories, but lean mass does.

CALCULATING TDEE (TOTAL DAILY ENERGY EXPENDITURE)

You multiply:

BMR × Activity Factor = TDEE

Activity factor depends on lifestyle:

LAZY PERSON EXAMPLE

- (Approx MET values)
- Sleeping (8–10 hrs) = MET 0.8
- Rest of day = MET 2 Average MET of day = 1.4
- Equivalent Activity Factor = ~1.4–1.5

ACTIVE PERSON EXAMPLE

- Sleeping (8–10 hrs) = MET 0.8
- Exercise (60–90 min) = MET 3.5–5
- Rest = MET 2
- Average MET = ~2.1
- Activity Factor = ~1.8–2.1

VERY ACTIVE

- Activity Factor ≈ 2.5
- Intensely Active
- Activity Factor ≈ 3.0–3.5
- Insanely Active
- Activity Factor ≈ 4.0+

WHICH METHOD SHOULD YOU USE? BMR OR TDEE?

If you have tendency to lose muscle easily →
Use TDEE deficit approach (slower, controlled fat loss)

If you don't lose muscle easily and fat loss is slow →
Use BMR-based deficit (more aggressive approach)

Understanding your own tendencies is essential.

TAKEAWAY

You must be fully aware of:

your true maintenance

your weekly averages

your calorie intake

your fluctuations

your BMR

your TDEE

Only then can you plan a correct deficit or surplus.

Being aware = being in control.